	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering (PCCOE) (An Autonomous Institute) Affiliated to Savitribai Phule Pune University (SPPU) ISO 21001:2018 Certified by TUV SUD	

PROJECT DIARY

Department: IT **Academic Year:** 2025–2026 **Year and Div.:** Final Year B

Title of Project: CipherDocs: A Secure AI-Powered Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification

Name and Sign of Internal Guide: Dr. Rohini Pise & Mr. Rohit Tate

Date / Duration	Main Topic	Work Done / Discussion Points	Mentor's Comments
Week 1 – Week 2 (January 2026)	**Phase 1: Problem Identification & Literature Survey**	Identified issues in traditional document verification systems such as forgery, centralized storage risks, delayed verification, and lack of transparency. Reviewed research papers on Blockchain, IPFS, AES encryption, SHA-256 hashing, smart contracts, and AI-based validation systems. Finalized the project scope, objectives, SDGs, and the title **"CipherDocs: Secure AI-Powered Blockchain-Based Document Verification System."**	Asked to review existing papers & review papers latest.
Week 3 – Week 4 (January 2026)	**Phase 2: Requirement Analysis & System Planning**	Identified functional and non-functional requirements. Finalized technologies including Python, Flask, Solidity, IPFS, Web3.py, AES-256, SHA-256, HTML/CSS/JavaScript, and Ganache. Planned the workflow for document upload, encryption, blockchain verification, AI validation, and decentralized storage. Designed the initial system architecture and workflow diagrams.	Discuss the objective & scope of project

Week 5 – Week 6 (February 2026)	**Phase 3: Dataset Collection & Data Preprocessing**	Collected sample academic certificates, IDs, and forged document samples for testing. Implemented metadata extraction, file validation, and duplicate checking. Applied SHA-256 hashing and AES-256 encryption for secure document handling. Prepared datasets and structured metadata for AI validation experiments.	Focus on size of documents
Week 7 – Week 9 (February – March 2026)	**Phase 4: Backend & Blockchain Development**	Developed REST API backend services for document upload, verification, encryption, and retrieval. Integrated IPFS for decentralized document storage and generated CID references. Created Solidity smart contracts for storing document hashes on the blockchain. Connected the blockchain with the backend using Web3.py and tested transaction consistency. Serialized AI models using Joblib into .pkl files for REST API deployment.	Satisfactory & advised to draft a paper for publication
Week 10 – Week 11 (March 2026)	**Phase 5: Frontend & Verification System Development**	Developed a responsive frontend dashboard for document upload and verification. Added blockchain transaction details, verification status, and document preview features. Implemented the complete workflow: upload → encryption → IPFS storage → blockchain anchoring → verification result display. Optimized UI responsiveness and overall user experience.	Make More interactive UI
Week 12 – Week 13 (March – April 2026)	**Phase 6: AI Validation & System Integration**	Integrated an AI-assisted validation module for tampered document detection and anomaly checking. Added an adaptive AES key rotation mechanism for enhanced security. Integrated frontend, backend, blockchain, IPFS, and AI modules into a unified framework. Performed real-time testing with multiple document formats and workflows.	Work on AI chatbot
Week 14 – Week 15 (April 2026)	**Phase 7: Testing & Performance Evaluation**	Conducted unit testing, integration testing, security testing, and blockchain validation testing. Verified tamper detection using regenerated hashes and blockchain comparison. Evaluated encryption accuracy, verification speed, and decentralized retrieval performance. Generated result graphs, workflow screenshots, and performance analysis reports.	Do more performance testing.

Week 16 (April 2026)	**Phase 8: Documentation & Final Deployment**	Prepared all black book chapters including Introduction, Literature Survey, Requirements, Methodology, Design, Implementation, Testing, Results, and Conclusion. Created UML diagrams such as Use Case, Class, Sequence, Activity, Component, DFD, and Deployment diagrams. Added screenshots, testing tables, architecture diagrams, and result analysis.	Fix some of format of documentation
Week 18 (First Week of May 2026)	**Phase 9: Final Review & Submission**	Completed the final review of the project report, PPT, and demo. Finalized the plagiarism report, GitHub repository, appendix, and user manual. Prepared the final deployment setup and demonstration workflow. Presented the research paper on **"CipherDocs"** at the IEEE CCGE 2026 conference and received the **Best Paper Award** in the respective track. Submitted the black book, source code, presentation, and supporting project documents.	Satisfactory Prepare & submit paper to any Conference

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